

Technical Challenge – Software Engineering

Context

A fictitious retail network needs to better understand its **sales**, **customers**, and **inventory**. Your mission will be to **model the ontology**, **process raw data**, **analyze** it, and **present business insights** in a React interface.

Challenge Steps

1) Ontology (Conceptual Model)

Build an **ontology diagram** that represents the main entities and relationships, inspired by Palantir Foundry models. Suggested entities:

- Customers
- Products
- Sales
- Inventory
- Categories

Relate:

- Who buys what, when, and for how much;
- Current inventory per product and turnover;
- Product categories and customer preferences.

2) Data Generation (CSV)

Generate **fictitious** datasets in CSV format (at least 3 files). You may use Faker, Python scripts, Mockaroo, or another tool.

- `customers.csv` → id, name, age, region
- `products.csv` → id, category, price, supplier
- `sales.csv` → id, customer_id, product_id, date, quantity

3) Data Pipeline in Python

Create an **ETL pipeline** that:

- Reads the CSVs;
- Handles inconsistencies (missing values, duplicates, outliers);
- Generates consolidated metrics (e.g.: revenue by region, top-selling products, customer churn).

4) Use of TypeScript

Implement a **TypeScript service** that:

- Consumes the processed data (via JSON file exported by the pipeline or API);
- Exposes endpoints for the frontend (e.g.: `/api/top-products`, `/api/sales-by-region`).

5) React Interface

Create at least **one screen** that:

- Shows two data visualizations (e.g.: sales by region chart, top-selling products ranking);
- Allows exploration/filtering (e.g.: by period, category, region);
- Is simple and clear (you may use Recharts, Victory, or Chart.js).

6) Business Insights

From the data, deliver at least **3 business insights** that support decision-making. Examples:

- Customers from the South region buy 30% more electronics than the national average;
- Product X represents 20% of revenue but is in critical stock;
- Young customers (<25 years old) concentrate purchases in the Fashion category.

Evaluation Criteria

- 1) **Ontology** – clarity and logic of the conceptual model;
- 2) **Data structuring and cleaning** – quality of the pipeline and documentation in Python;
- 3) **Coding and interface** – best practices in Python + TypeScript + React;
- 4) **Creativity and business insights** – relevance and decision-making potential.